Case 44: WU-E heat-parameter response

1 Verification purpose

This case verifies the response of the intended downscaled Wildland–Urban ELMFIRE (WU–E) model-2 heat calculation to the design-fire curve, target building-fuel properties, heat-release-rate (HRR) ellipse adjustment, the radiation cutoff, and multiple simultaneous sources. It is designed to expose breakpoint errors, confusion between a plateau duration and its absolute end time, use of the wrong target-property path, incorrect heat-area normalization, missing cutoff logic, a failure to accumulate transient heat, or nonlinear source aggregation.

Nineteen deterministic simulation conditions use one or two stationary WU-E sources. Nonburnable fire behavior fuel model (FBFM) 93 background and a zero spread-adjustment raster suppress propagation; spotting, terrain, and weather variation are disabled. The intended downscaled formulation is the numerical acceptance reference. The reviewed configuration interpretation, spread-rate calculation, and level-set formulation were identical to the specified ELMFIRE source revision; this identity claim does not extend to all other model components. Differences between the simulated response and the intended equations below are reported as deviations.

Verification question. Do all simulation conditions reproduce the complete time-dependent heat-release rate per unit area (HRRPUA), direct-flame-contact (DFC), radiation, and accumulated-heat fields within 0.005 normalized L_1 error, while satisfying the exact parameter scalings, radiation support, source-superposition, and no-propagation invariants?

Current result: PASS. 19 of 19 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

For elapsed burning time t, the implementation reads four values from the source building-fuel-model row: growth endpoint t_e, absolute plateau endpoint t_f, decay endpoint t_d, and peak HRRPUA H. It evaluates

$$\dot{q}''_{\mathrm{HRR}}(t) = \begin{cases} Ht/t_e, & t \le t_e, \\ H, & t_e < t \le t_f, \\ H(t-t_d)/(t_f-t_d), & t_f < t \le t_d, \\ 0, & t > t_d. \end{cases} \tag{1}$$

The fixed wind and geometry determine the UCB ellipse once per source. With linear characteristic building dimension $L_c = \texttt{BLDG_AREA}$ in metres, source area $A_s = L_c^2$, ellipse major and minor radii B and A_e, and $c = \texttt{HRR_ELLIPSE_ADJ}$, the intended spatial normalization is

$$C_h = \frac{A_s}{\pi(cB)(cA_e)}, \qquad c_{default} = 1. \tag{2}$$

For target cell j, define $\beta_{ssd} = \min[1, (s_{sd} + L_c)/\Delta x]$ and select $K_\theta = B + E$, $B - E$, or $2A_e$ for downwind, upwind, or sidewind sectors. The clipped DFC coverage is $f_d = \mathrm{clip}[(K_\theta + \Delta x/2 - r\beta_{ssd})/\Delta x, 0, 1]$. Radiation uses $r_{eff} = r\beta_{ssd}c$, physical $HAZ = 100$ m, and the complementary factor $(1 - f_d)$. Spatial nonburnable fraction n_j takes precedence when supplied; the building-model/default value is the fallback. The table supplies absorptivity α_j. Each source s contributes

$$q''_{\text{DFC},s,j} = g_f \eta(W) 0.65(1-n_j) f_{d,s,j}\, \dot{q}''_{\text{HRR},s}(t) C_h, \tag{3}$$

$$q''_{\text{rad},s,j} = g_f \eta(W) \frac{0.35(1-n_j)\alpha_j f_{r,s,j} \dot{q}''_{\text{HRR},s}(t) A_s}{4\pi r^2_{\text{eff},s,j}}, \tag{4}$$

$$\eta(W) = \frac{3.25}{1+\exp[-0.05(W-75)]} + 0.5, \qquad g_f = \min[(\Delta x/L_c)^2, 1]. \tag{5}$$

Radiation overlap is exactly zero outside the physical heat-affecting zone, with half-cell clipping. Transient maps sum all burning sources. Total received heat is updated every one-second solver step as

$$Q_j^{\text{total}} \mathrel{+}= q''_j(t)\, \Delta t\, \Delta x^2. \tag{6}$$

The independent case-specific Python reference calculation implements these equations directly; it imports no ELMFIRE code and has no execution source-tree dependency. The pinned source currently omits η, uses 0.3 rather than 0.35 radiation, omits the 0.65 DFC factor and grid corrections, and uses cell area in C_h.

3 Derivation of expected behavior

The four source curves deliberately sample every breakpoint. The base curve uses $(t_e, t_f, t_d, H) = (4, 8, 12, 100)$ in SI units; the peak simulation condition doubles H, the short curve uses $(2, 4, 6, 100)$, and the long-plateau curve uses $(4, 10, 14, 100)$. Because every heat term is linear in HRRPUA, doubling H must double the HRR and both heat-flux maps at every dump.

At fixed geometry, DFC scales with spatial $1 - n_j$, while radiation scales with $(1 - n_j)\alpha_j$. The spatial input therefore predicts exact response ratios for $n_j \in \{0, 0.2, 0.5, 1\}$ and $\alpha_j \in \{0, 0.4, 0.8, 1\}$. DFC must not change in the absorptivity sweep. At fixed B and A_e, both heat fields scale as c^{-2} for $c \in \{0.25, 0.5, 0.75, 1\}$.

The sum from a left and a right source must equal the two-source result cell by cell and dump by dump. Likewise, the final accumulated rasters must equal the sum of every one-second transient raster times $\Delta t \Delta x^2$. Finally, two simulation conditions change the spatial nonburnable-fraction raster from zero to one while retaining the same building-fuel-model table. The fully nonburnable target must receive zero DFC and radiation; equality would expose the current source's table-only behavior.

The peer-reviewed WU–E papers support the transient design-fire, target heat, and multi-source coupling concepts [1, 2]. The detailed formulation above is stated explicitly. The ends of growth, fully developed burning, and decay are specified as absolute transition times.

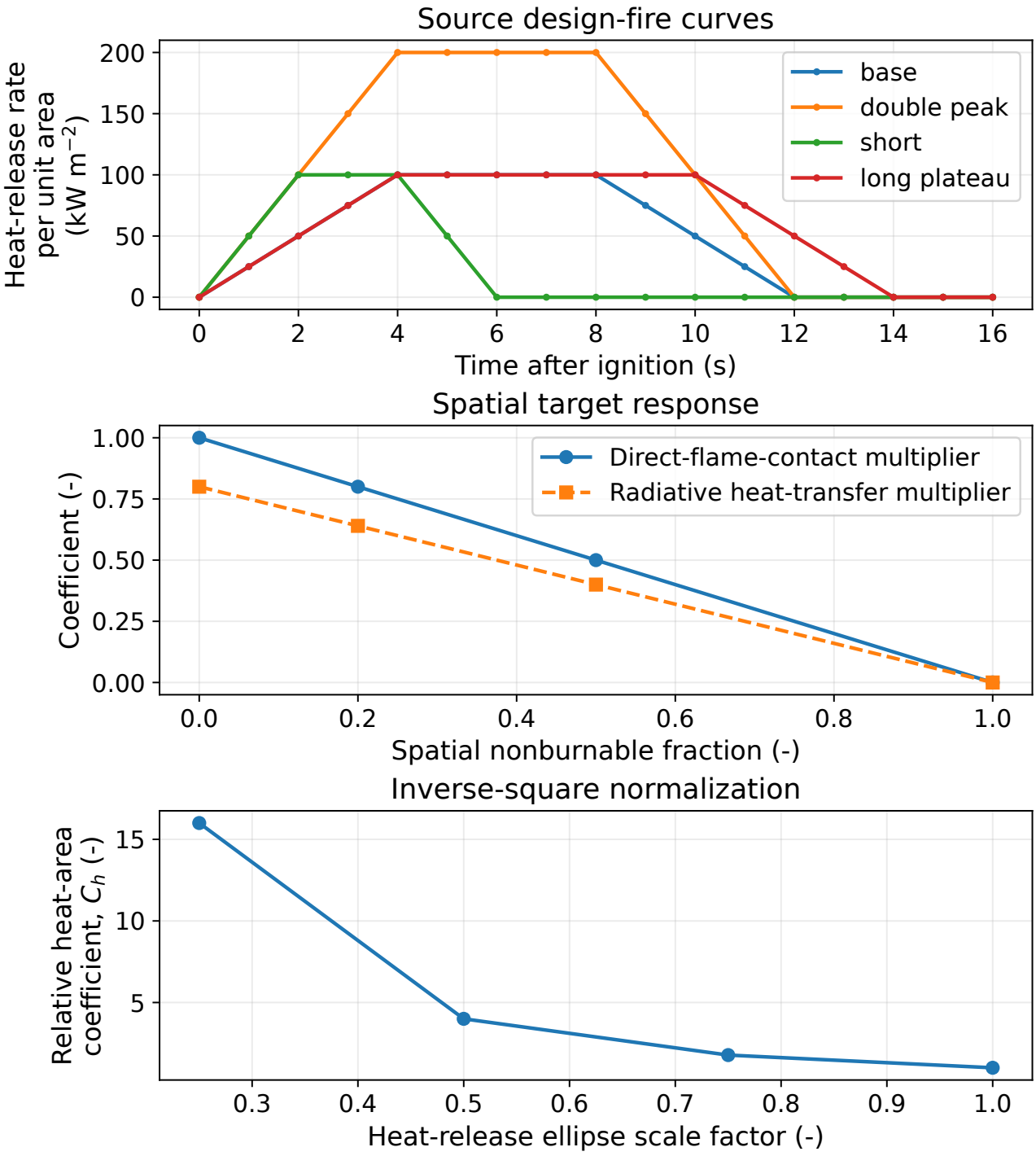


Figure 1: Independent predictions for the HRR curve families, spatial target coefficients, and inverse-square ellipse normalization. These curves are generated before, and independently of, an ELMFIRE run.

4 Verification metrics and acceptance criteria

For nonzero reference array R and output array E, the principal metric is

$$\epsilon_{L_1} = \frac{\sum |E - R|}{\sum |R|}. \tag{7}$$

The sums cover all 16 scheduled dump times and every raster cell. A zero-reference array uses absolute L_1 error, so a supposedly suppressed component cannot pass through a zero denominator. Nodata, non-finite values, unexpected shapes, duplicate result fields, and output times other than exactly 1 through 16 seconds are rejected. The prescribed HRR curves include $t = 0$, but ELMFIRE's first scheduled output occurs after one scheduled output interval. A confirmed successful completion must have matching SHA-256 identities for the simulation settings, spatial inputs, both fuel-property tables, reference and analysis methods, ELMFIRE program, simulation record, and all results. Output labels alone are not sufficient evidence. The relationship between the identified model program and the stated source revision must be established separately.

Expected property	Observable and metric	Criterion	Failure sensitivity
HRR breakpoints	Full HRR raster stack versus reference calculation	$\epsilon_{L_1} \leq 0.005$	timing, curve row, peak
Spatial heat fields	Full DFC/radiation stacks versus reference calculation	each $\epsilon_{L_1} \leq 0.005$	coefficients, cutoff, kernel
Exact parameter response	Scaled-pair raster errors	each $\leq 10^{-5}$	spatial n_b, absorptivity, or c^{-2}
Time accumulation	Transient sums versus final totals	$\epsilon_{L_1} \leq 0.005$	update/dump ordering
Source aggregation	Pair versus sum of two singles	$\leq 10^{-5}$	aggregation defect
Isolation	time of arrival (TOA) and off-source HRR	zero extra cells	unintended propagation

Overall PASS requires every metric and all 19 simulation conditions. Incomplete, stale, ambiguous, or malformed evidence yields NOT EVALUABLE, never PASS.

5 Simulation configuration and predicted outcome

Parameter	Configured value	Isolation role
Grid	55 by 55, 20 m cells	contains full update band
CRS	EPSG:32610	explicit metric coordinates

Time	$\Delta t = 1$ s; stop 16 s; dump 1 s	exact breakpoints/integration
Wind	12 m/s toward raster east	fixed ellipse geometry
Sources	one center or two separated cells	single/superposition pair
Background	FBFM93 with zero adjustment	prevents propagation
Spotting/terrain	disabled/flat	removes confounders
Model selector	WUI model type 2	UMD/UCB source path

The point ignition is scheduled at 17 seconds, just beyond the 16-second stop. It prevents generic stalled-front termination without adding a physical source during the test. All input rasters are regenerated with common dimensions, CRS, transform, nodata value, and explicit data types.

The predicted result is a stationary source set whose heat maps follow each source curve, vanish beyond the implementation's support, add linearly, and integrate exactly into the final received-heat fields.

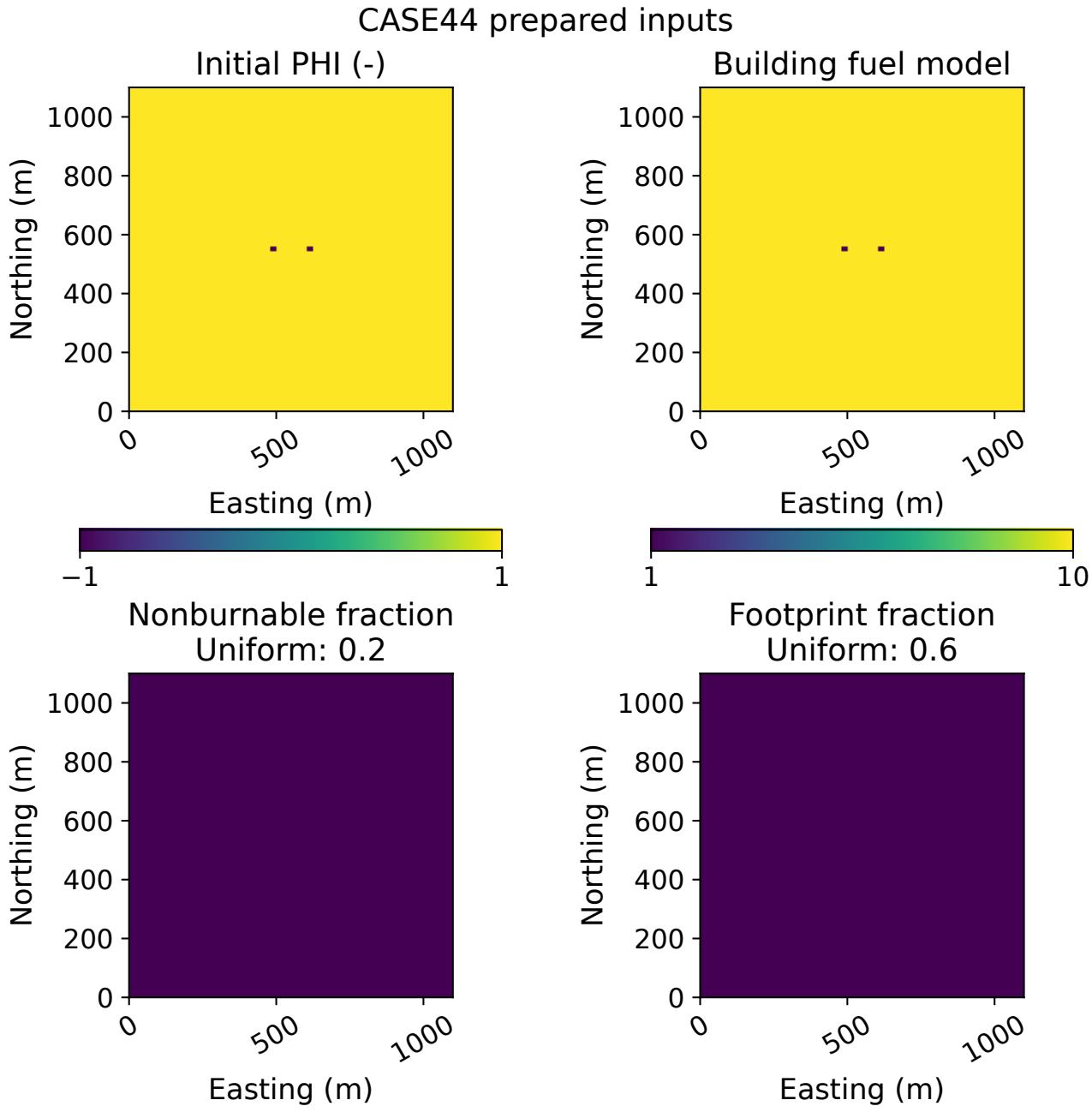


Figure 2: Prepared two-source configuration read back from generated GeoTIFFs. Source model IDs override the uniform target-table ID only at the two initially burning cells.

6 Actual simulation results

The current analysis procedure state is **INCOMPLETE**; the scientific decision is **NOT EVALUABLE**, with 0 of 19 required simulation conditions having evaluable execution evidence. Reason: Metrics source data failed the report consistency contract.. The common executed-binary SHA-256 is −−.

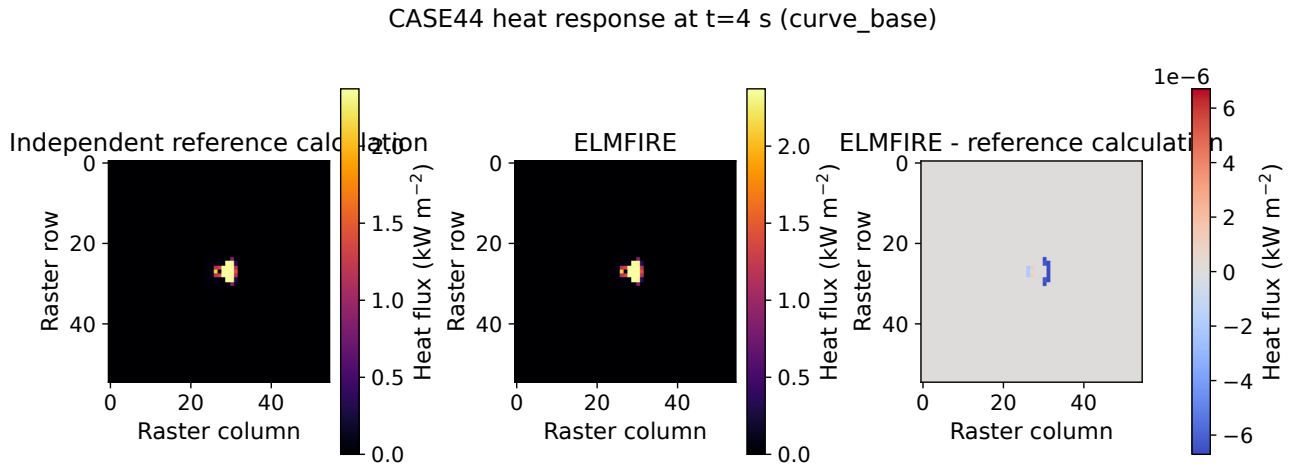


Figure 3: Representative whole-domain ELMFIRE result, independent reference calculation, and difference. This figure is generated only from completed output.

7 Calculated metrics and verification decision

Maximum HRR curve normalized L1 error — PASS

Criterion	$\leq$ 0.005
Observed	*Units: fraction.* 0

Double-peak HRR scaling error — PASS

Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 0

Maximum DFC full-stack normalized L1 error — PASS

Criterion	$\leq$ 0.005
Observed	*Units: fraction.* 1.68513e-06

Maximum radiation full-stack normalized L1 error — PASS

Criterion	$\leq$ 0.005
Observed	*Units: fraction.* 7.0351e-07

Spatial nonburnable scaling error — PASS

Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 8.05946e-08

Target-table absorptivity scaling error — PASS

Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 8.05946e-08

DFC Heat-release ellipse scale factor inverse-square error — PASS

Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 1.10746e-07

Maximum radiation outside source cutoff — PASS

Criterion	$\leq$ 1e-06
Observed	*Units: kW m^-2.* 0

Maximum transient-to-total integration error — PASS

Criterion	$\leq$ 0.005
Observed	*Units: fraction.* 6.72838e-08

Two-source superposition error — PASS	
Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 1.30613e-10
Spatial nonburnable-raster response — PASS	
Criterion	fully nonburnable / combustible heat $\leq$ 1e-05
Observed	*Units: ratio.* 0
Maximum arrival-mask mismatch count — PASS	
Criterion	0
Observed	*Units: cells.* 0
Maximum unexpected HRR source count — PASS	
Criterion	0
Observed	*Units: cells.* 0

Current decision: NOT EVALUABLE. NOT RUN is a analysis procedure state, not evidence of verification and never maps to PASS.

8 Technical interpretation

The required transient stacks are unavailable because the solver stops when this one- or two-source setup has too few tagged propagation cells. The dormant ignition controls the separate PHI-stall test, not this low-node stopping rule. An overrun final dump does not contain the missing rise, plateau, and decay history, so the current decision remains NOT EVALUABLE.

Once complete histories exist, the comparisons separate distinct causes: HRR errors indicate curve timing or source-state problems; correct HRR with incorrect heat maps indicates transfer coefficients, support, or normalization; a transient-to-total mismatch indicates accumulation or dump timing. The spatial-nonburnable and source-superposition controls test their respective input and aggregation paths without substituting planned values for output.

9 Scope and limitations

This case verifies deterministic source HRR and heat-transfer mechanics for a fixed model-2 ellipse. It does not verify physical calibration, flame-shape regressions across wind and geometry (CASE43), ignition thresholds, rate of spread (ROS), wildland–urban transitions, embers, fuel consumption, or convergence. Its intended spatial-property interpretation is tested against the pinned source revision. The reference calculation is evaluated independently of the ELMFIRE simulation.

References

[1] D. M. J. Purnomo et al., “Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model,” *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.